

SMART INDIA HACKATHON 2024



Problem Statement Details

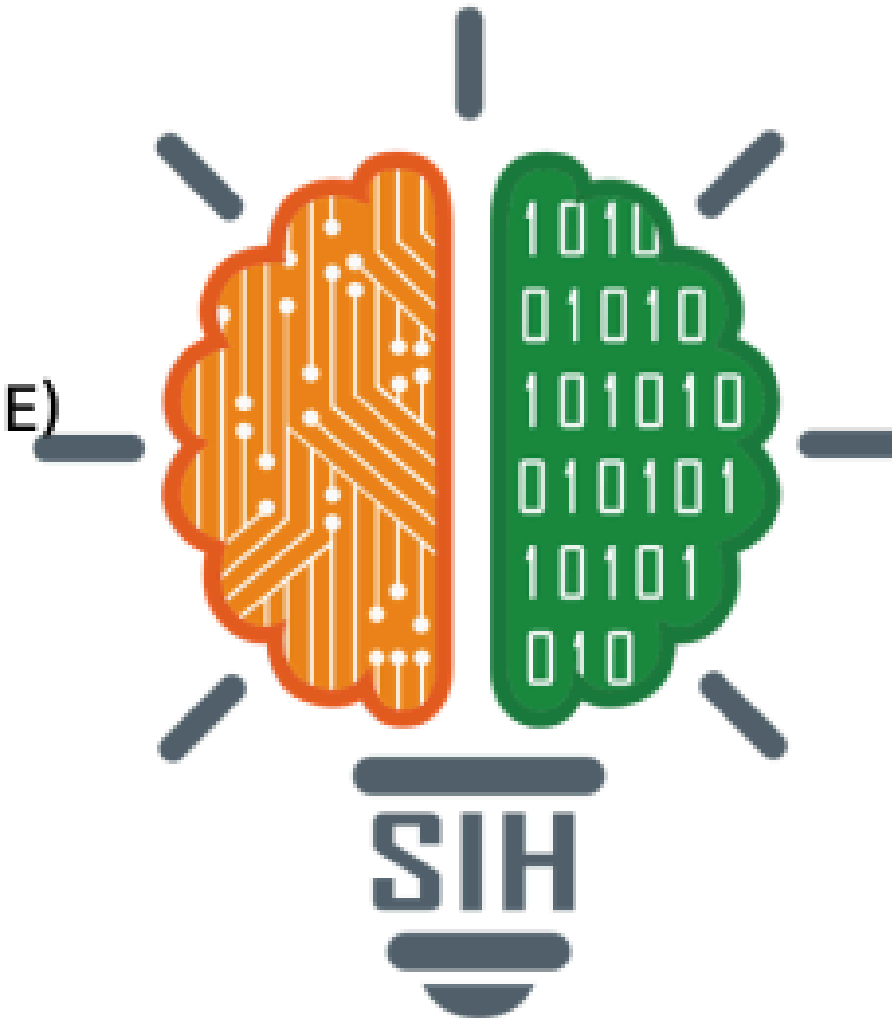
- Problem Statement ID – 1777
- Problem Statement Title-
- **Theme-** Smart Education
- **PS Category** Software
- **Organization :** Ministry of Skill Development and Entrepreneurship (MSDE)

Team Details

Collage : Davv, Indore
Institute name : IET-Davv

Team ID - 33954

Team Name - Future Frameworks





SkillVista is an AI-driven assessment tool that offers comprehensive, inclusive evaluations across various formats—pen-paper, online MCQs, practical exams, and viva voce. It adapts to individual needs, supports PWD candidates with accessibility features, and provides real-time feedback and analytics.

Key features of Our AI-Powered Assessment Tool

Demo Website Link: <https://sih-2-nine.vercel.app/>

Holistic Assessment Coverage

Covers a wide spectrum of skills across various educational levels, from schools to vocational training.

Inclusivity for PWD Candidates

Designed with accessibility features to ensure fair assessments for Persons with Disabilities.

Multi-Mode Assessment

Supports online, offline, and blended modes, catering to diverse candidate environments

AI-Powered Personalization

Utilizes AI to adapt assessments based on individual performance and learning needs

Real-Time Analytics and Feedback

Provides detailed insights to candidates and educators for continuous improvement.

Standardization and Quality Assurance

Ensures consistent quality and rigor in assessments across different regions and institutions.

Data Privacy and Security

Implements strong measures to protect candidate data and ensure confidentiality.

Unique Features

A 24/7 available customized AI chatbot that help user with general queries and doubts also help them to explore the web site

GitHub Repo : <https://github.com/Black-Jade0/AI-Models>

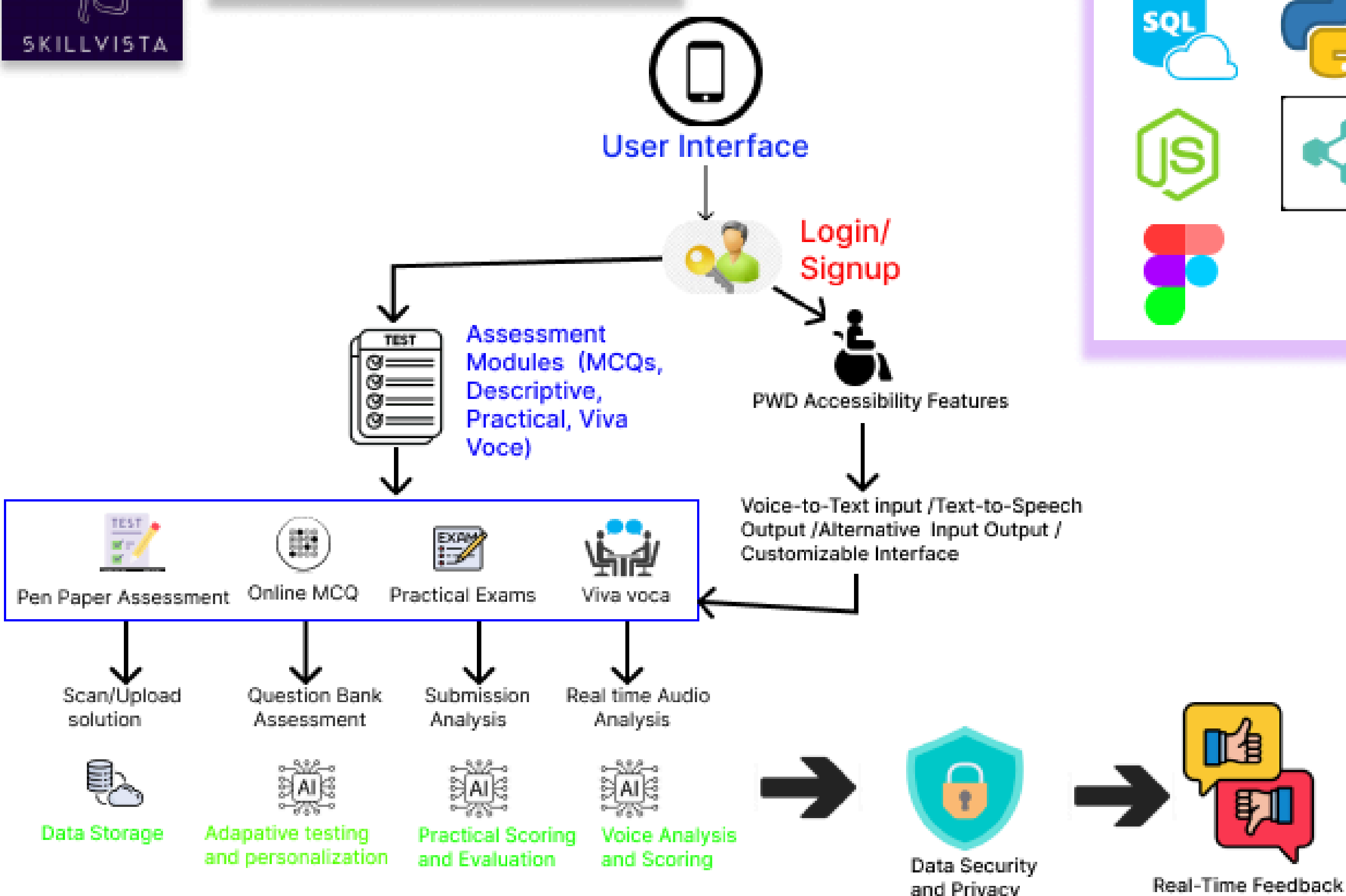
This repository contains

- AI driven **Practical exam assessment evaluation** model using keras_ocr and sentence transformer ("all-mpnet-base-v2")
- AI driven **speech to text** and **text to speech** models for PWD students using wishper AI , google sr and gTTS.
- AI driven **real time eye tracking** for online assessments

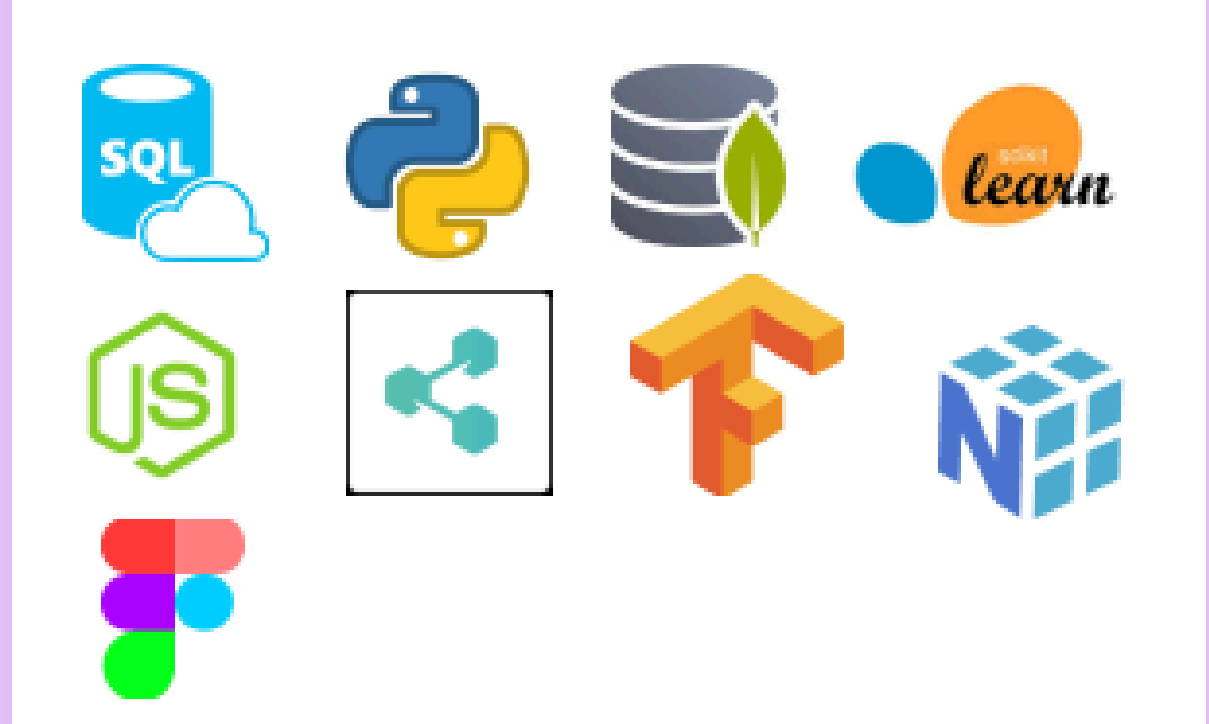
24/7 ChatBot : <https://cdn.botpress.cloud/webchat/v2/shareable.html?botId=0dc1fc0e-650b-4f49-85eb-f30bf5492a0c>



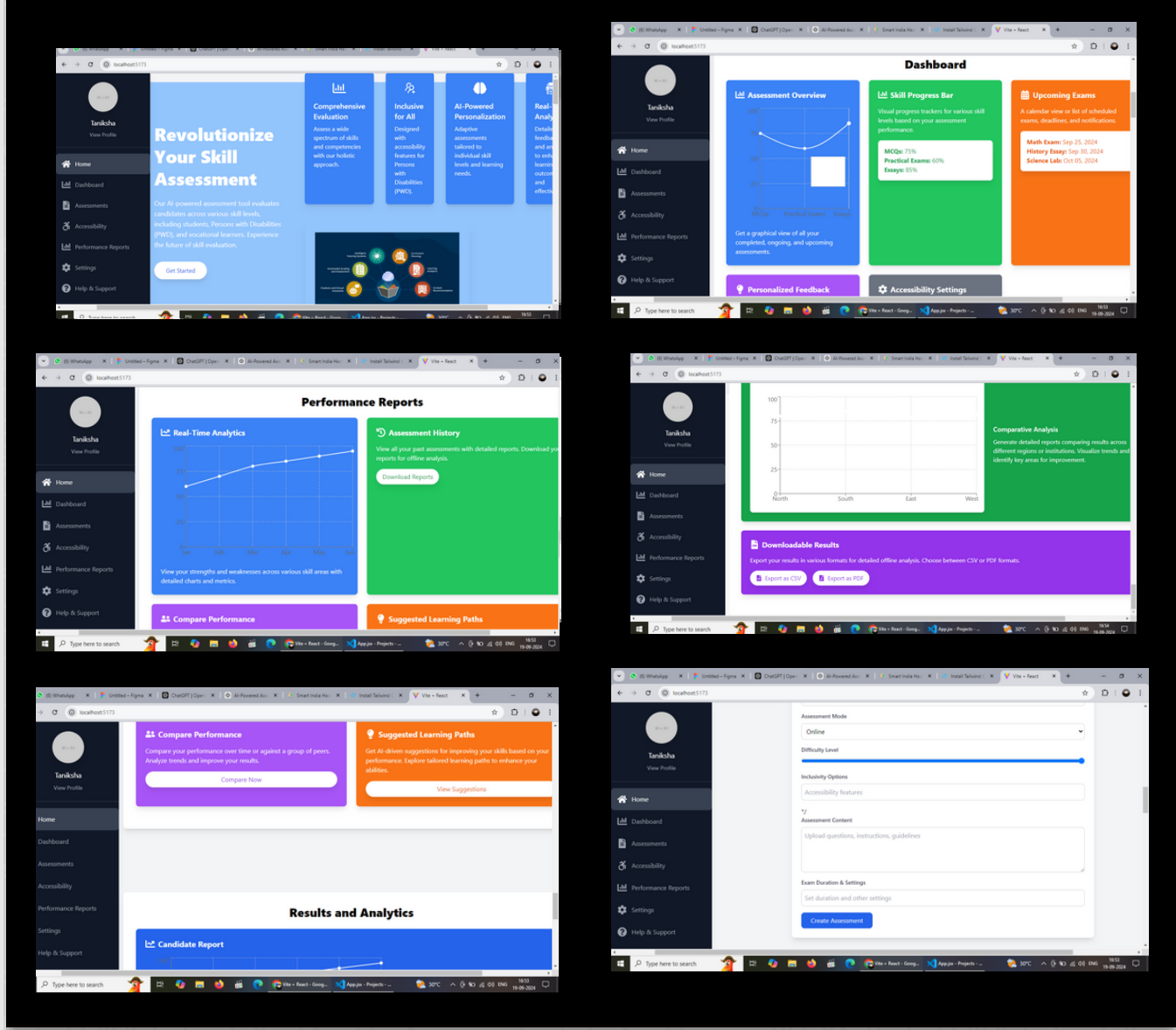
Technical Approach



Tech Stack



Frontend of our website:





Feasibility and Viability

Feasibility Analysis

- 1. Mature AI Technologies:** Adaptive testing, practical scoring, and voice analysis are well-established and can be integrated using existing AI frameworks.
- 2. Accessible Tools for PWD:** Technologies like voice-to-text, text-to-speech, and alternative input methods are widely available, making inclusivity achievable.
- 3. Multi-Mode Assessment:** Online, offline, and blended assessments are feasible using cloud-based platforms with offline syncing capabilities.
- 4. Scalable Infrastructure:** Cloud computing ensures scalability, allowing the tool to support large numbers of candidates across diverse regions.
- 5. Regulatory Compliance:** Data privacy and security standards like GDPR and India's Data Protection Bill provide clear frameworks for safe data handling.

Viability Analysis

- 1. Market Demand:** The diverse Indian skill ecosystem and growing emphasis on inclusive education create a strong demand for a standardized assessment tool.
- 2. Government Support:** Initiatives like Digital India and Skill India align with this platform, increasing the likelihood of adoption and funding.
- 3. Cost-Effective:** Leveraging cloud services and existing AI models reduces development and operational costs, making the solution affordable for institutions.
- 4. User-Friendly Technology:** Accessible, low-barrier technologies for PWD ensure ease of use and encourage broad adoption.
- 5. Competitive Advantage:** A tool combining multi-mode, AI-driven, and PWD-inclusive assessments can offer a unique value proposition in the education and training sectors.



Impacts and Benefits

Key Benefits of the AI-Powered Assessment Tool for Skill Evaluation



Improved Access to Assessments

Enables candidates from remote and underserved areas to participate in assessments through online, offline, and blended modes.



Increased Inclusivity

PWD candidates will have equal opportunities to demonstrate their skills with accessible, adaptive assessments.



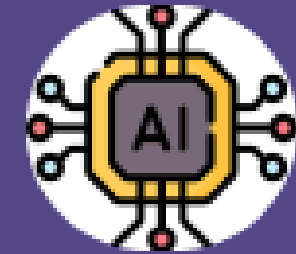
Enhanced Learning Outcomes:

Real-time feedback and personalized assessments lead to improved performance and targeted learning for all candidates.



Increased Flexibility:

The tool's ability to function online, offline, or in blended modes offers unparalleled flexibility for institutions and candidates.



Efficiency and Scalability:

AI automation streamlines the assessment process, reducing manual intervention, improving efficiency, and making the system easily scalable.

ShowStopper

Our platform provides a 24/7 chat support for the students for their post assessment doubts / general queries and instruction related to the website



SKILLVISTA

Research and References

AI in Special Education: AI tools can significantly improve inclusivity in education by offering tailored support for students with disabilities. Technologies such as text-to-speech (TTS), speech-to-text (STT), and predictive text tools enhance accessibility for students with visual, physical, or learning disabilities, allowing more flexible and personalized assessments. This is crucial for adapting assessment methods like online exams and viva voce for students with varying needs([Ciddl](#))([Information Technology Services Support](#)).

Inclusivity for Persons with Disabilities (PWD): AI-based learning platforms such as adaptive learning systems provide dynamic and personalized learning paths. These platforms adjust to the student's ability level in real time, enabling students with disabilities to progress at their own pace while still engaging in assessments designed for the broader skill ecosystem([Information Technology Services Support](#)).

Accessibility Tools in Assessment: For PWD students, AI can facilitate inclusive assessments by automatically generating captions, converting text to Braille or audio formats, and translating content into different languages. These functionalities ensure that assessments like MCQs, practicals, and viva voce are accessible to all students([Ciddl](#))([Information Technology Services Support](#)).

Data Analytics for Feedback and Adaptation: AI-powered real-time analytics can identify trends and challenges faced by students with disabilities during assessments, offering immediate feedback to educators. This data helps institutions to adapt assessments for fairness and inclusivity while maintaining rigor([Ciddl](#))

Microsoft Research on Accessible AI: Microsoft's research outlines design principles for creating AI systems that are inclusive for PWDs. It emphasizes accessibility features such as voice-to-text, text-to-speech, and alternative input methods, all of which are critical to ensuring equitable assessment platforms([Microsoft](#)).

AI Fairness for PWD: A study on AI fairness highlights that fairness for people with disabilities differs significantly from other protected attributes, requiring personalized approaches. It explores how AI systems can be inclusive, with a focus on handling sensitive data and avoiding biases, particularly for PWD, emphasizing fairness across different disability manifestations([ar5iv](#)).

Team Details:

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